



1.6 Proof by induction (HL)

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Proof by mathematical induction establishes a statement $P(n)$ for all integers $n \geq n_0$ in three steps: (1) Base case — verify $P(n_0)$ is true; (2) Inductive step — assume $P(k)$ is true for some $k \geq n_0$, and show this assumption implies $P(k+1)$ is true; (3) Conclusion — since $P(n_0)$ holds and truth passes from k to $k+1$, $P(n)$ is true for all integers $n \geq n_0$ by the principle of mathematical induction.

Worked example: follow the reasoning

Prove that $1+2+3+\dots+n = n(n+1)/2$ for all $n \geq 1$.
Base case $n=1$: LHS=1, RHS= $1(2)/2=1$. True.
Assume true for $n=k$: $1+2+\dots+k = k(k+1)/2$.
For $n=k+1$: $1+2+\dots+k+(k+1) = k(k+1)/2 + (k+1) = (k+1)(k/2+1) = (k+1)(k+2)/2$, which matches the formula with $n=k+1$.
Since true for $k=1$ and $P(k) \Rightarrow P(k+1)$, the statement is true for all $n \geq 1$ by induction.

Quick check · multiple choice

In a proof by induction, what must be assumed in the inductive step?

- A That the statement is true for $n=1$ only
- B That the statement is true for all n
- C That the statement is true for some $n=k$
- D That the statement is false for $n=k$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Prove by mathematical induction that $3^n > n^2$ for all integers $n \geq 2$.

[7]

Final answer: _____



Answers and explanation

Question 1 [7]

Base case $n=2$: $3^2=9 > 4=2^2$. True.

Assume true for $n=k$ ($k \geq 2$): $3^k > k^2$.

For $n=k+1$: $3^{(k+1)} = 3 \cdot 3^k > 3k^2$ (using assumption).

Need to show $3k^2 \geq (k+1)^2 = k^2 + 2k + 1$ for $k \geq 2$, i.e. $2k^2 - 2k - 1 \geq 0$, which holds for $k \geq 2$ (at $k=2$: $8 - 4 - 1 = 3 \geq 0$, and increasing).

So $3^{(k+1)} > 3k^2 \geq (k+1)^2$, giving $3^{(k+1)} > (k+1)^2$.

Since true for $n=2$ and $P(k) \Rightarrow P(k+1)$, by induction $3^n > n^2$ for all integers $n \geq 2$.

Quick check answer

C. The inductive step assumes $P(k)$ is true for some particular k , then proves $P(k+1)$ follows.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.